



Week 3 Lecture 6

▼ Class	BSCCS2003
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🔗 Materials	
# Module #	21
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Tools (part 2)

Programmatic HTML generation: PyHTML

- Composable functions - each function generates a specific output
- Example:
 - To generate an `h1` heading: the function should return ..

```
"<h1>text of heading</h1>"
```

```
import pyhtml as h

t = h.html(
    h.head(
        h.title('Test page')
    ),
    h.body(
        h.h1('This is a title'),
        h.div('This is some text'),
        h.div(h.h2('inside title'),
            h.p('some text in paragraph'))
    )
)
print(t.render())
```

More complex HTML

```
def f_table(ctx):
    return (
        tr(
            td(cell) for cell in row
```

```
) for row in ctx['table']  
)
```

Templates

- Standard template text
- Placeholder / Variables
- Basic (very limited) programmability
- Examples:
 - Python inbuilt string templates - good for simple tasks
 - Jinja2 - used by Flask
 - Genshi
 - Mako
 - ...

Jinja

- Ties in closely with Flask
- Template functionality with detailed API

Remember: Templates can generate any output, not just HTML

```
from string import Template  
  
t = Template('$name is the $job of the $company')  
s = t.substitute(name='Tim Cook', job='CEO', company='Apple Inc.')  
print(s)
```

Jinja example

```
from jinja2 import Template  
t = Template('Hello {{ something }}!')  
print(t.render(something='world'))  
  
t = Template('My favourite numbers: {% for n in range(1, 10) %}{{ n }} ' '{% endfor %}')
```